

# Maja Lujan Niemeyer

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## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>PhD in Physics</b>                                                                                           | final grade: summa cum laude (highest distinction) |
| Max Planck Institute for Astrophysics, Garching                                                                 | October 2021 - February 2026                       |
| <b>M.Sc. in Physics</b>                                                                                         | final grade (best possible grade: 1.0): 1.16       |
| Thesis: <i>Lyman-<math>\alpha</math> Halos around High-Redshift Lyman-<math>\alpha</math> Emitting Galaxies</i> | grade: 1.0                                         |
| Ludwig-Maximilians-Universität München                                                                          | April 2019 - September 2021                        |
| <b>B.Sc. in Physics</b>                                                                                         | final grade (best possible grade: 1.0): 1.6        |
| Thesis: <i>Sky Subtraction Techniques for Multiplexed Integral Field Spectrographs</i>                          | grade: 1.0                                         |
| Ludwig-Maximilians-Universität München                                                                          | October 2017 - January 2019                        |
| Georg-August-Universität Göttingen                                                                              | October 2016 - September 2017                      |

## RESEARCH INTERESTS

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Line intensity mapping, Lyman- $\alpha$  radiative transfer, reionization, low-surface brightness phenomena

## FIRST-AUTHOR PUBLICATIONS

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TOTAL CITATIONS: 82

- $\diamond$  **M. Lujan Niemeyer**, E. Komatsu et al. (2025)  
*Ly $\alpha$  Intensity Mapping in HETDEX: Galaxy-Ly $\alpha$  Intensity Cross-Power Spectrum*  
Astrophysical Journal 999 177, arXiv:2510.11427  
Contribution: full analysis, modeling, and interpretation of the measurement and writing of the draft.
- $\diamond$  **M. Lujan Niemeyer** (2025)  
*Effect of Lyman- $\alpha$  Radiative Transfer on Intensity Mapping Power Spectra*  
Astrophysical Journal 980 250, arXiv:2407.03060
- $\diamond$  **M. Lujan Niemeyer**, J. L. Bernal, E. Komatsu (2023):  
*SIMPLE: Simple Intensity Map Producer for Line Emission*  
Astrophysical Journal 958 4; arXiv:2307.08475  
Contribution: developed, tested, and published the code; wrote the draft.
- $\diamond$  **M. Lujan Niemeyer**, W. P. Bowman, R. Ciardullo et al. (2022):  
*Ly $\alpha$  Halos around [O III]-selected Galaxies in HETDEX*  
Astrophysical Journal Letters 934 L26; arXiv:2207.11098  
Contribution: full analysis and interpretation of the measurement and writing of the draft.
- $\diamond$  **M. Lujan Niemeyer**, E. Komatsu, C. Byrohl et al. (2022):  
*Surface Brightness Profile of Lyman- $\alpha$  Halos out to 320 kpc in HETDEX*  
Astrophysical Journal 929 90; arXiv:2203.04826  
Contribution: full analysis and interpretation of the measurement and writing of the draft.

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**OTHER PUBLICATIONS**


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- ◇ E. Mentuch Cooper, K. Gebhardt, D. Davis et al. (including **M. Lujan Niemeyer**, 2026)  
*HETDEX Public Data Release 1: Source Catalog 2 and Data Cubes from 90 sq deg of Integral-Field Optical Spectroscopy*  
 Astrophysical Journal Supplement Series 284 67; arXiv:2606.04208
  
- ◇ E. Mentuch Cooper, K. Gebhardt, D. Davis et al. (including **M. Lujan Niemeyer**, 2026)  
*Ly $\alpha$  Nebulae in HETDEX: The Largest Statistical Census Bridging Ly $\alpha$  Halos and Blobs across Cosmic Noon*  
 Astrophysical Journal 1000 38; arXiv:2603.06410
  
- ◇ M. Mirza Khanlari, K. Gebhardt, L. H. Weiss et al. (including **M. Lujan Niemeyer**, 2025)  
*The HETDEX Survey: Probing Neutral Hydrogen in the Circumgalactic Medium of 88,000 Lyman Alpha Emitters*  
 Astrophysical Journal 989 169; arXiv:2507.15942  
 Contribution: helped write the discussion; edited the draft.
  
- ◇ L. H. Weiss, K. Gebhardt, D. Davis et al. (including **M. Lujan Niemeyer**, 2025)  
*Using Ly $\alpha$  Absorption to Measure the Intensity and Variability of  $z \sim 2.4$  Ultraviolet Background Light*  
 Astrophysical Journal 983 72; arXiv:2504.13253  
 Contribution: created analysis, code, and text for Figure 3; provided ideas for the interpretation; edited the draft.
  
- ◇ L. H. Weiss, D. Davis, K. Gebhardt et al. (including **M. Lujan Niemeyer**, 2024):  
*Absorption Troughs of Lyman Alpha Emitters in HETDEX*  
 Astrophysical Journal 962 102; arXiv:2401.02490  
 Contribution: provided and tested several ideas to confirm that this is real signal; edited the draft.
  
- ◇ D. Davis, K. Gebhardt, E. Mentuch Cooper et al. (including **M. Lujan Niemeyer**, 2023)  
*HETDEX Public Source Catalog 1—Stacking 50,000 Lyman Alpha Emitters*  
 Astrophysical Journal 954 209; arXiv:2307.03096  
 Contribution: provided ideas for the analysis; edited draft.
  
- ◇ E. Mentuch Cooper, K. Gebhardt, D. Davis et al. (including **M. Lujan Niemeyer**, 2023)  
*HETDEX Public Source Catalog 1: 220 K Sources Including Over 50 K Ly $\alpha$  Emitters from an Untargeted Wide-area Spectroscopic Survey*  
 Astrophysical Journal 943 177; arXiv:2301.01826  
 Contribution: tested the public data product; contributed to the HETDEX API; edited the draft.
  
- ◇ C. Liu, K. Gebhardt, E. Mentuch Cooper et al. (including **M. Lujan Niemeyer**, 2022)  
*The Active Galactic Nuclei in the Hobby–Eberly Telescope Dark Energy Experiment Survey (HETDEX) II. Luminosity Function*  
 Astrophysical Journal 935 132; arXiv:2207.11801  
 Contribution: edited the draft.

- ◇ K. Gebhardt, E. Mentuch Cooper, R. Ciardullo et al. (including **M. Lujan Niemeyer**, 2021)  
*The Hobby–Eberly Telescope Dark Energy Experiment (HETDEX) Survey Design, Reductions, and Detections*  
 Astrophysical Journal 923 217; arXiv:2110.04298  
 Contribution: developed the full-frame sky subtraction technique; edited the draft.
- ◇ D. Davis, K. Gebhardt, E. Mentuch Cooper et al. (including **M. Lujan Niemeyer**, 2021)  
*Detection of Lyman Continuum from  $3.0 < z < 3.5$  Galaxies in the HETDEX Survey*  
 Astrophysical Journal 920 122; arXiv:2107.11319  
 Contribution: wrote the code for and the explanation of the weighted biweight location estimator; edited the draft.

## TALKS

- ◇ Fachbeirat (external advisory board) Seminar at Max Planck Institute for Astrophysics March 2026
- ◇ Large-Scale Structure Days 2025 October 2025
- ◇ Line Intensity Mapping 2025 June 2025
- ◇ HETDEX Collaboration Meeting 2025 May 2025
- ◇ Line Intensity Mapping 2024 June 2024
- ◇ HETDEX Collaboration Meeting 2024 May 2024
- ◇ Alpine Cosmology Workshop July 2023
- ◇ Present and Future of Line Intensity Mapping Conference April 2023
- ◇ **Invited:** The Multiphase CGM Conference February 2023
- ◇ What Matter(s) around Galaxies conference September 2022
- ◇ Institute seminar at Max Planck Institute for Astrophysics May 2022
- ◇ **Invited:** seminar at Max Planck Institute for Extraterrestrial Physics March 2022

## RESPONSIBILITIES AND SERVICE

- ◇ **Deputy Chair** of the HETDEX Cosmology Science Working Group February 2026 - now
- ◇ **Co-Investigator** of the VIRUS Deep Field project 2020 - now

## OUTREACH AND TEACHING

- ◇ **Café und Kosmos** outreach talk 2026
- ◇ **Podcast Interview at Bayerischer Rundfunk:**  
 invited guest in a live national TV and radio podcast about young scientists in Germany 2025
- ◇ **MPA planetarium:** explained the research conducted at MPA to > 200 students 2022 - now
- ◇ **Girl's day at MPA:** helped organize outreach event for female high-school students 2023
- ◇ **Mentored** a new PhD student through the MPA buddy system 2022
- ◇ **Outreach Article:** MPA Research Highlight 2022
- ◇ **Official Blog** of the University of Göttingen: wrote articles about studying physics 2017
- ◇ **Teaching Assistant** at University of Göttingen (Mathematical Methods for Physicist) 2017

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## VISITING RESEARCH AND SCHOOLS

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- ◇ University of Texas at Austin February - March 2024
- ◇ University of Texas at Austin February - March 2020
- ◇ University of Texas at Austin February - September 2019
- Research Assistant: Developed a new sky subtraction algorithm for multiplexed integral-field spectrographs to optimize the data processing for line intensity mapping measurements.
- ◇ First Light School on Stars, Galaxies, and Black Holes in the Epoch of Reionization
- Advanced School, IAG/University of São Paulo, Brazil July - August 2019

## SKILLS

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|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Python</b>             | very good skills; experience with, e.g., astropy, numpy, scipy, matplotlib, cython, numba, dask, mpi4py, multiprocessing, sklearn, emcee, cobaya. |
| <b>C/C++</b>              | good skills; experience with OpenMP                                                                                                               |
| <b>LaTeX</b>              | very good skills                                                                                                                                  |
| <b>Linux command line</b> | basic skills                                                                                                                                      |
| <b>Git</b>                | basic skills                                                                                                                                      |
| <b>HPC facilities</b>     | TACC, MPCDF computing facilities                                                                                                                  |
| <b>Hydrodynamic codes</b> | Athena, Arepo, and Gadget                                                                                                                         |
| <b>Languages</b>          | German (native), English (fluent), Spanish (basics), Latin (Latinum)                                                                              |

## CITIZENSHIP

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Germany, USA

## REFERENCES

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Eiichiro Komatsu

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Eric Gawiser

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